



Vehicle Content Optimization (VCO) at General Motors

Peiling Wu-Smith, Ph.D., Research Manager and Technical Fellow, Advanced Analytics Center of Expertise

Phil Keenan, Ph.D., Principal Technical Fellow, Advanced Analytics Center of Expertise

Jon Owen, Ph.D., Director of Advanced Analytics Center of Expertise, Chief Scientist for AI/ML and Operations Research

Andy Norton, Executive Director of Global Market Research

Kelly Kamm, Director of Global Vehicle and Technology Planning

Kathryn Schumacher, Ph.D., Staff Researcher, Advanced Analytics Center of Expertise

Peter Fenyes, Ph.D., Staff Researcher (retired), Advanced Analytics Center of Expertise

Don Kiggins, Features & Technology Content Optimization Manager, Global Market Research

Phi Konkel, Data Science Fellow, Advanced Analytics Center of Expertise

Bill Rosen, Technical Director (retired), Global Marketing Research

Kurt Schmitter, Staff Researcher, Advanced Analytics Center of Expertise

Sharon Sheremet, Director of Global Product Research & Insights, Global Market Research

Laura Yochim, Content Optimization Analyst, Global Market Research

Submission for the IISE Outstanding Innovation in Service Systems Engineering Award

GM Vehicles Have Hundreds of Features...



Vehicle Content

Vehicle Order Guides Define Contenting Decisions (what Features are offered as standard or optional content)



S: Standard content

A: Available as an upgrade for an additional charge

Blank: Not available in this trim

	Low Trim	Mid Trim	High Trim
Rear Vision Camera	S	S	S
Rear Park Assist		A	A
Perforated Leather-Appointed Heated Seats			S
8-Way Power-Adjustable Driver's Seat		A	S
LTZ Exterior Appearance			S
Convenience Package		A	
Outside Heated Power-Adjustable Mirrors			
Automatic Climate Control		A	S
Power Sunroof		A	S
Fog Lamps, Front Halogen			A
Driver Confidence Package		A	A
Universal Home Remote		A	A

Importance of Developing Optimized Order Guides



- The product life-cycle in automotive is much longer than in most industries.
- Billions of dollars spent each year in designing new vehicles and bringing them to market.
- The vehicle development process is long and complex
- Contenting decisions are made 2-4 years in advance and locked down
- Drive downstream decisions on engineering development, tooling investment, material sourcing, and manufacturing planning
- Decisions made years in advance can swing profitability by hundreds of millions of dollars and significantly impact our customers' experiences.

Traditional Approach



“Rearview”



Looked back at historical sales to see how well similar vehicles sold in the past

“Sideview”



Benchmarked against competitor offerings and imitated what competitors do



A Forward-Looking “Windshield” View

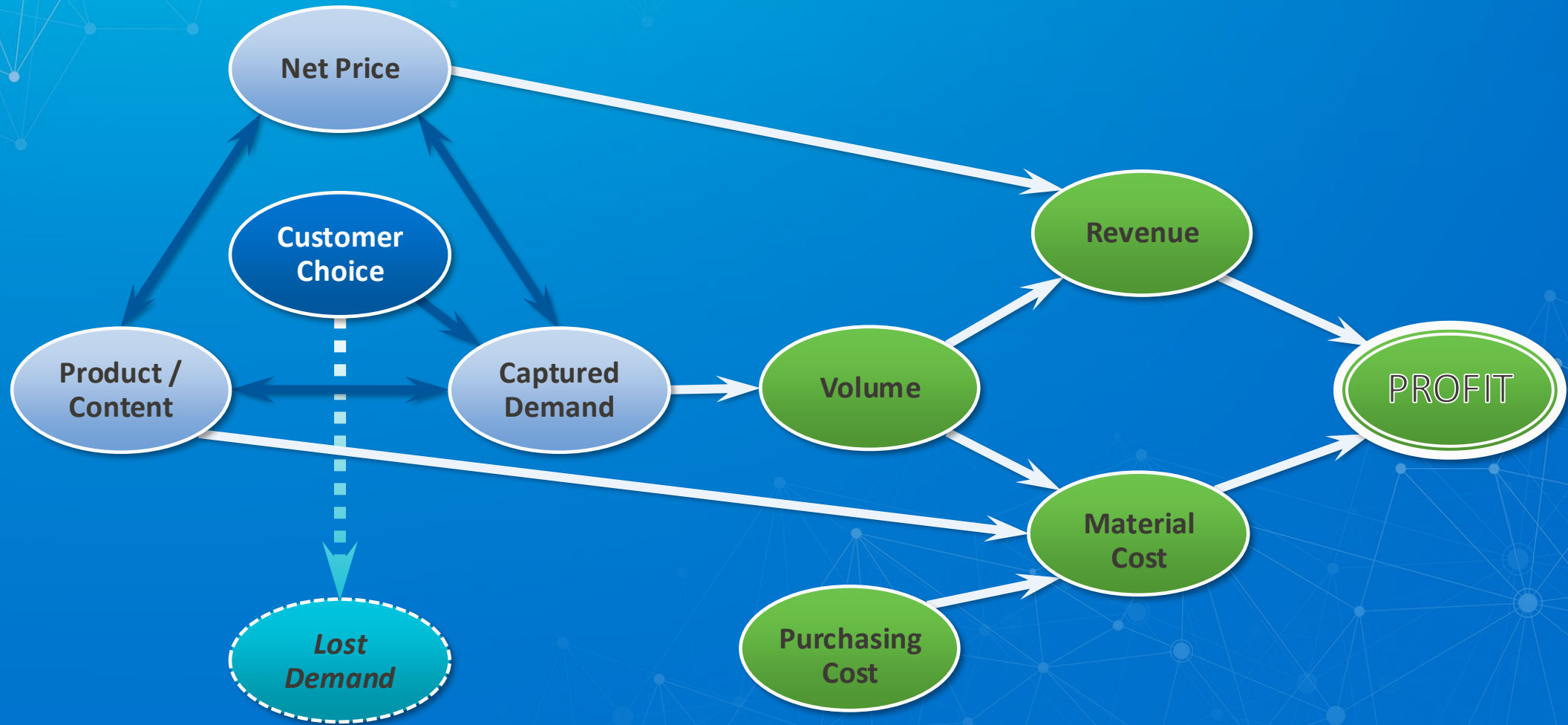
- “Rearview”: past success or failure does not tell you about changing consumer preferences, untapped opportunities, or interest in new technologies.
- “Sideview”: mirroring competitive offerings at best allows GM to be a “fast follower”.
- A “windshield view” helps us better understand the diversity of what consumers really want.



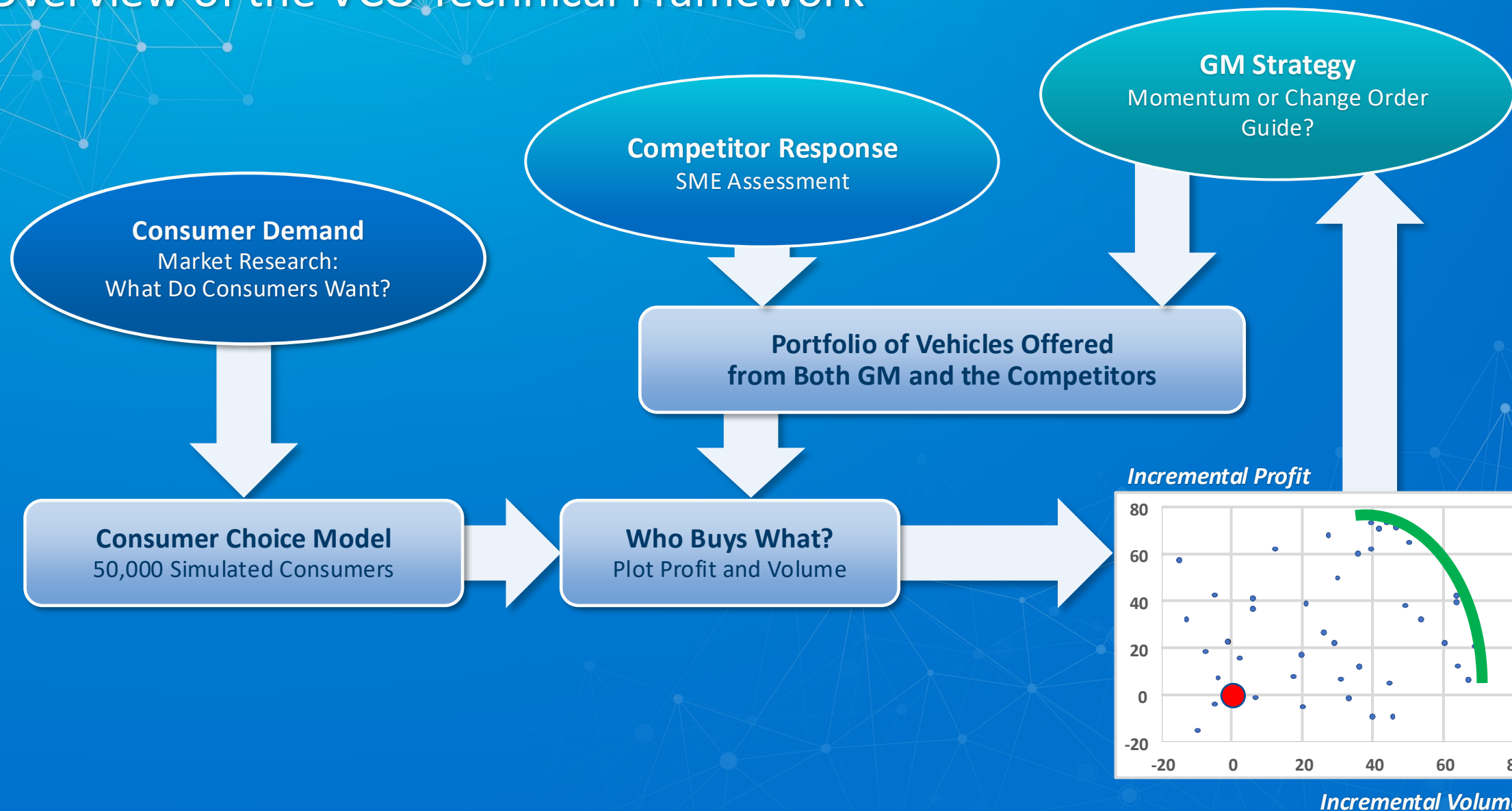
...Our Approach

- ✓ Forward-looking, customer-centric
- ✓ Comprehends complex dynamics & tradeoffs
- ✓ Optimizes vehicle contenting plan for win-win
 - ✓ Better satisfy customers
 - ✓ Improve profitability

Key to VCO: Modeling Dynamics of Consumer Choice and Comprehending Complex Dynamics and Tradeoffs



Overview of the VCO Technical Framework



Surveying Customers About 100+ Vehicle Features is a Big Challenge



One innovation:

- We combine conjoint with other types of market research tasks, each providing a different lens.

SafetyInteriorPowertrain and Vehicle DynamicsExteriorInfotainment

Safety (General)
The following features are already selected because they are standard features of the vehicle:

☒ 6 Airbags; Safety Belts 3-Point with Pre-Tensioners, all positions; Rear Back-up Camera - \$0

☒ 5 Star Government Safety Rating, (Overall Rating) - \$0

Airbags
Please select one of the following:

☒ 6 airbags - \$0

☐ 7 airbags - \$75

☐ 8 airbags - \$350

☐ 10 airbags - \$500

Cruise Control
Please select one of the following:

☒ Cruise Control - \$0

☐ Cruise Control with Low Speed Collision Avoidance - \$300

☐ Adaptive Cruise Control with Full Speed Range Collision Avoidance - \$1195

Parking Assist
Please select one of the following:

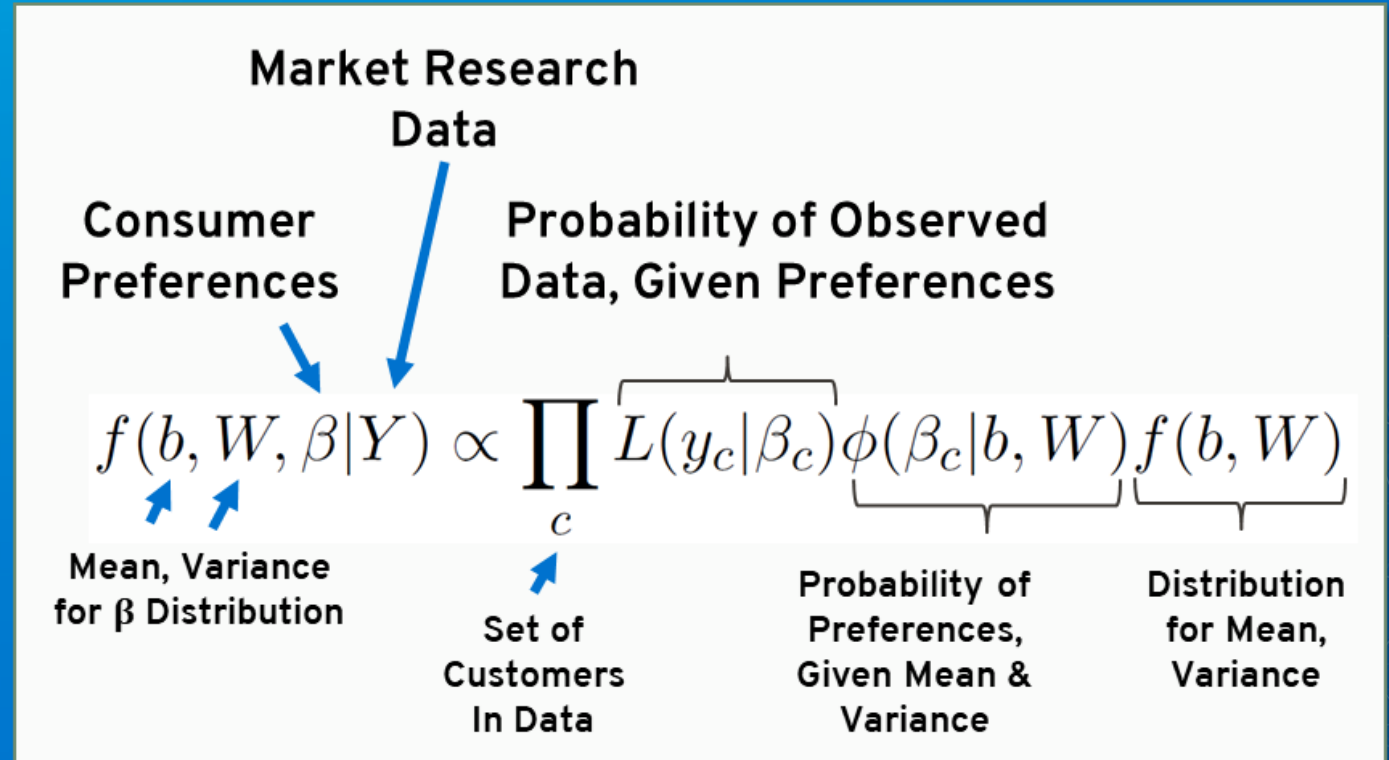
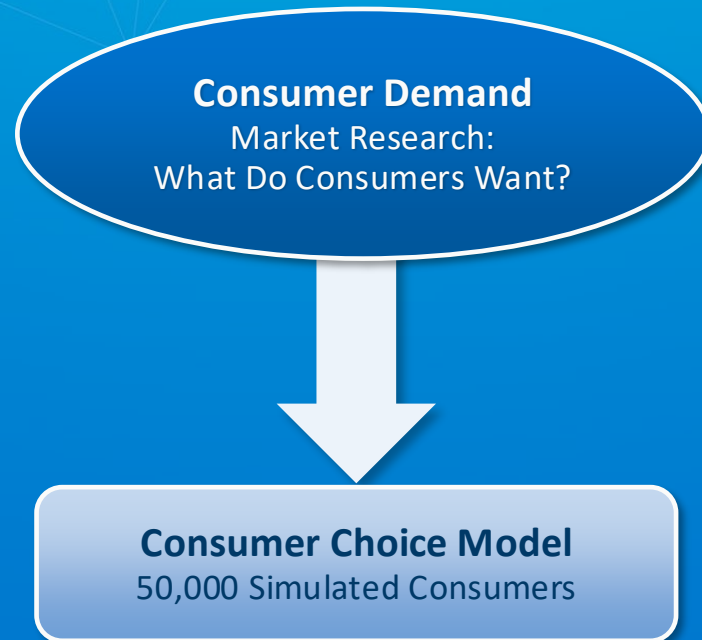
☒ No Parking Assist - \$0

☐ Rear Parking Assist - \$295

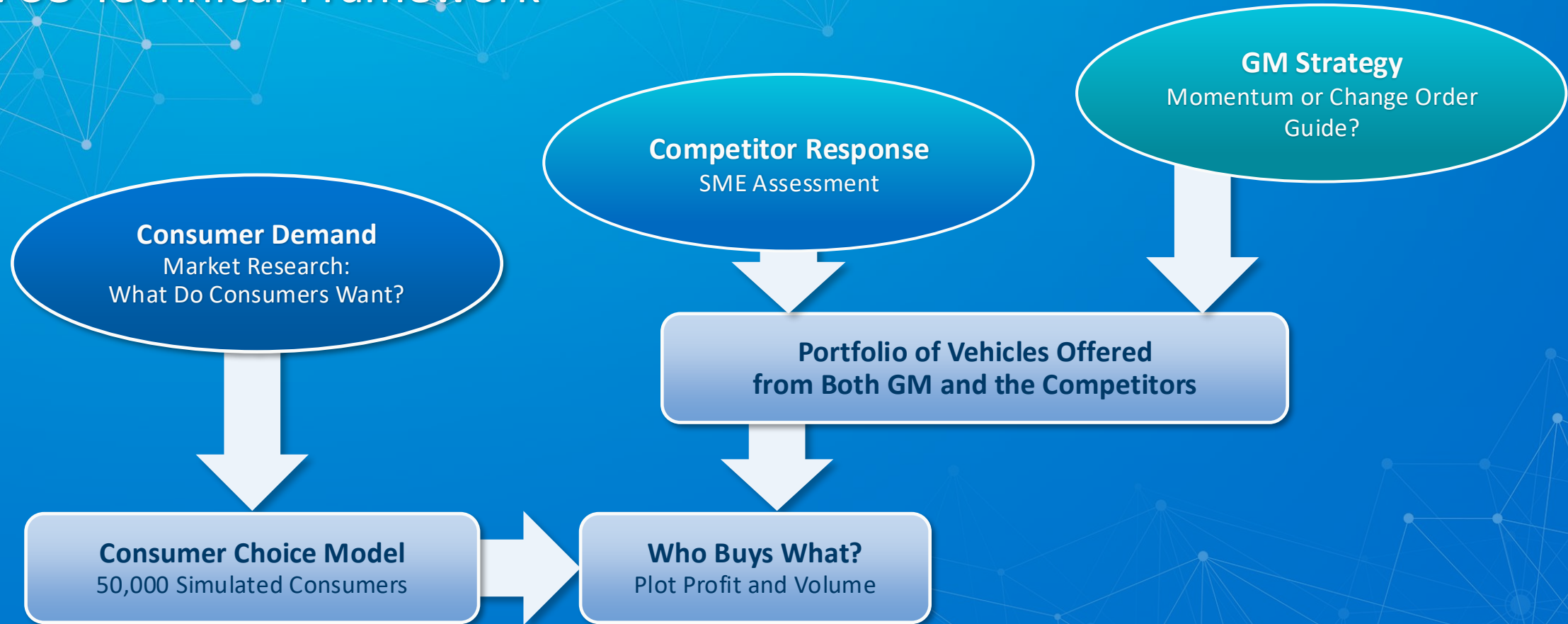
☐ Front and Rear Parking Assist - \$500

Base price: \$ 18,000Added options: \$ 0Total Price: \$ 18,000

VCO Technical Framework



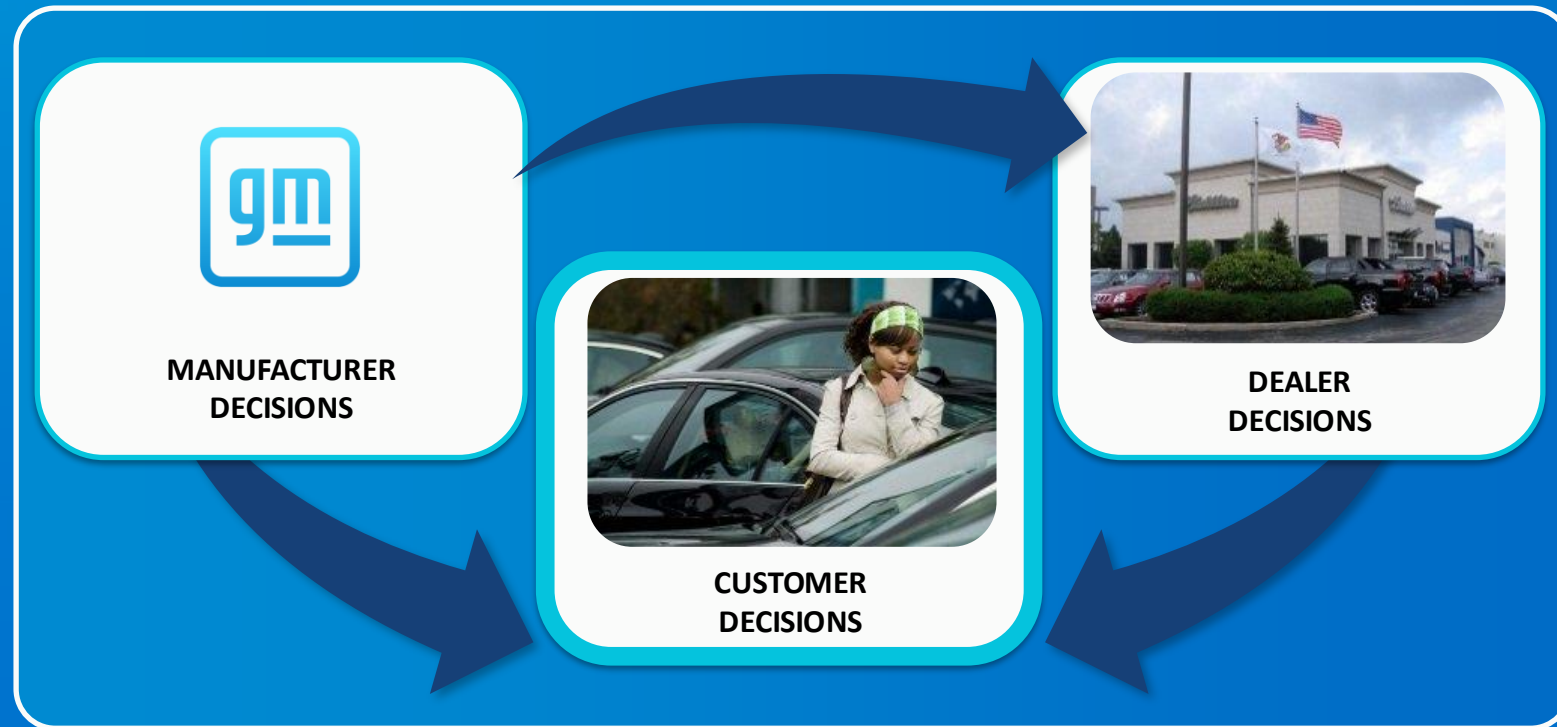
VCO Technical Framework



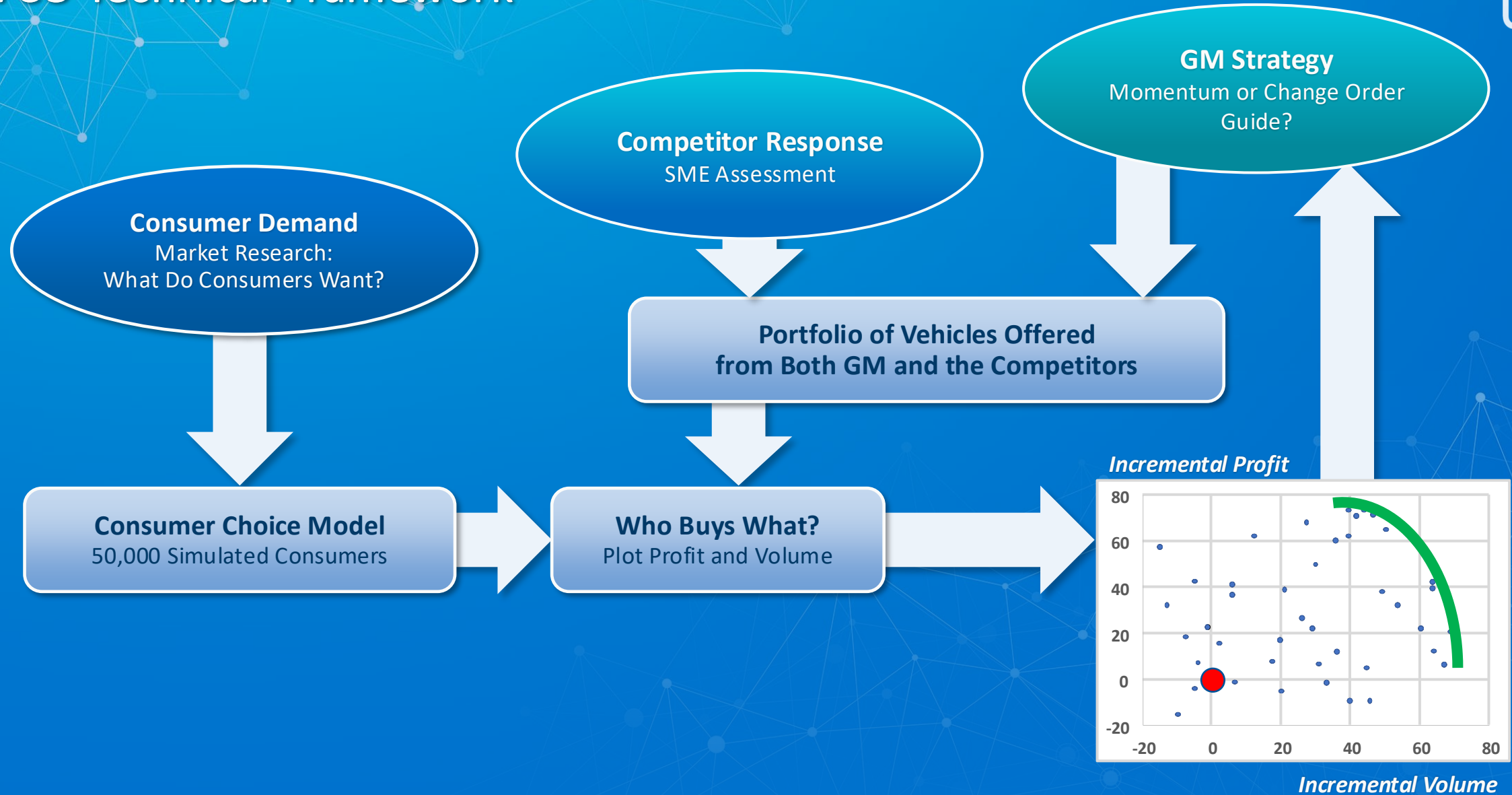
We Invented Several Enhancements to a Traditional Mixed Logit Model to Address Practical VCO Issues



- Independence of Irrelevant Alternatives (red bus / blue bus)
- Selling vehicles from dealer inventory
- Consumer budget constraints
- Non-compensatory choice behavior
- Estimating competitor response
- Calibrating to historical sales, time on lot and price elasticity data

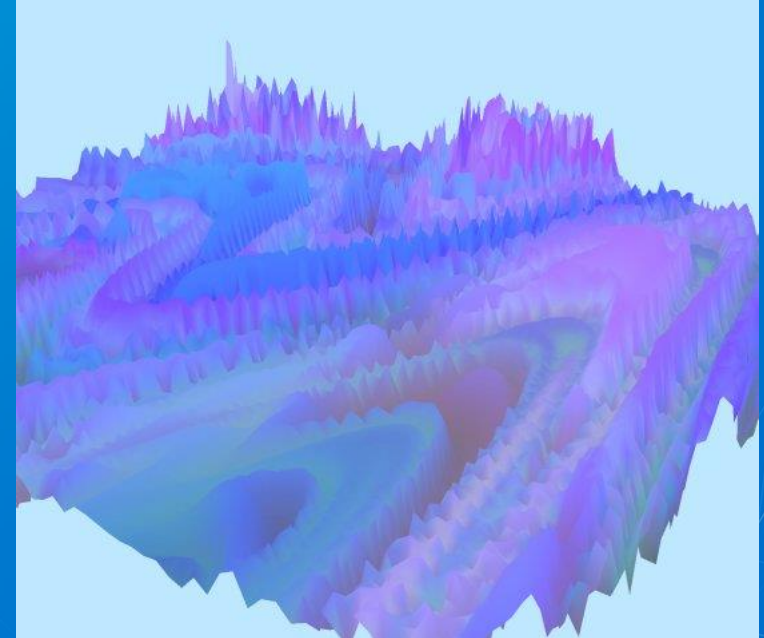


VCO Technical Framework



Objective Function is Non-linear and Multi-modal

- Driven by individual-level choice model enhanced from mixed logit
- What is optimal for one customer may be very different than what is optimal for another: creates multi-modal surface



$$\max \sum v(\beta, p) r(p)$$

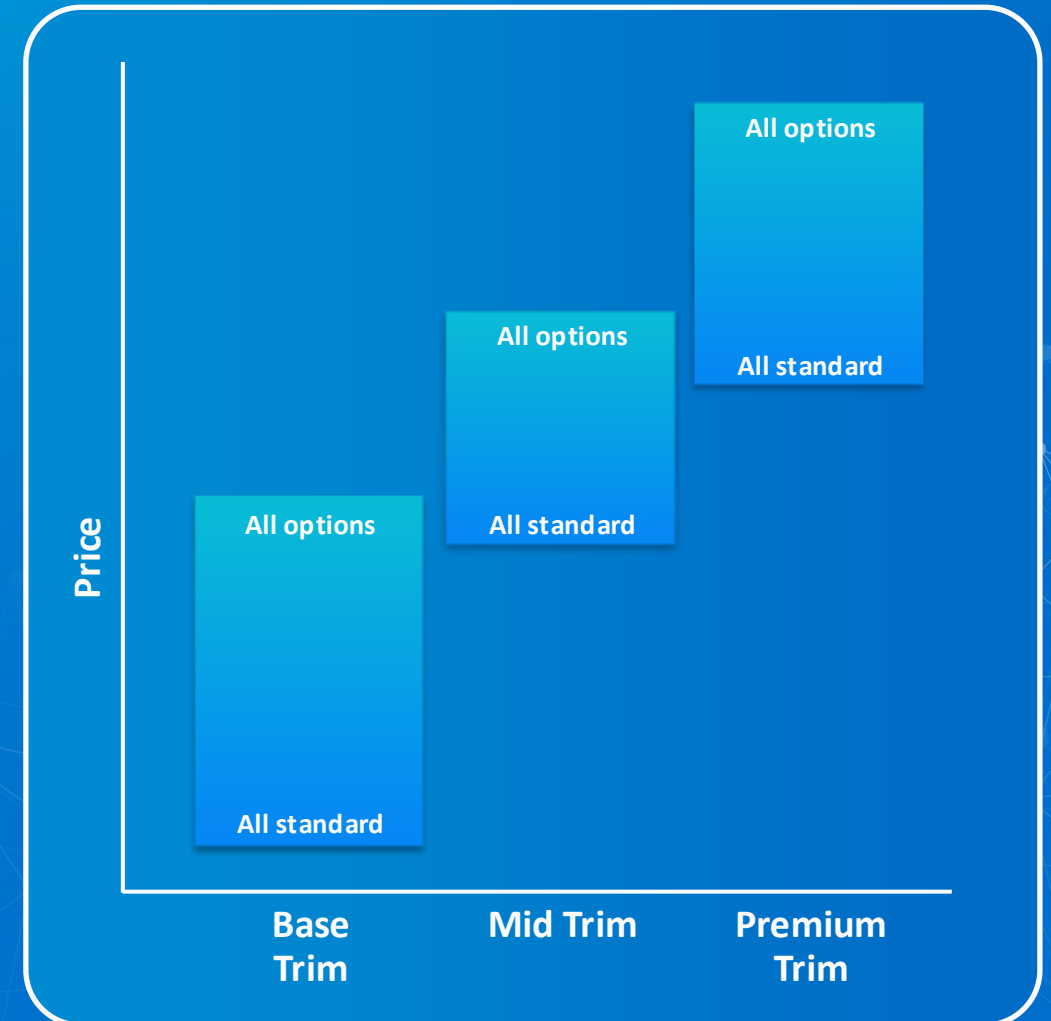
Volume: non-linear function
of individual customer
preferences, product
attributes

Variable profit:
function of prices,
variable costs



Feasible Solutions Must Satisfy Business Constraints

- Trim ladder: “higher” trims have more and better content than lower ones
- Engineering restrictions:
 - A vehicle with feature X must also have feature Y, and cannot have Z
- Price constraints:
 - The “all standard” in the Low trim must be cheaper than the “all standard” in the Mid trim
- And many more...



Optimization



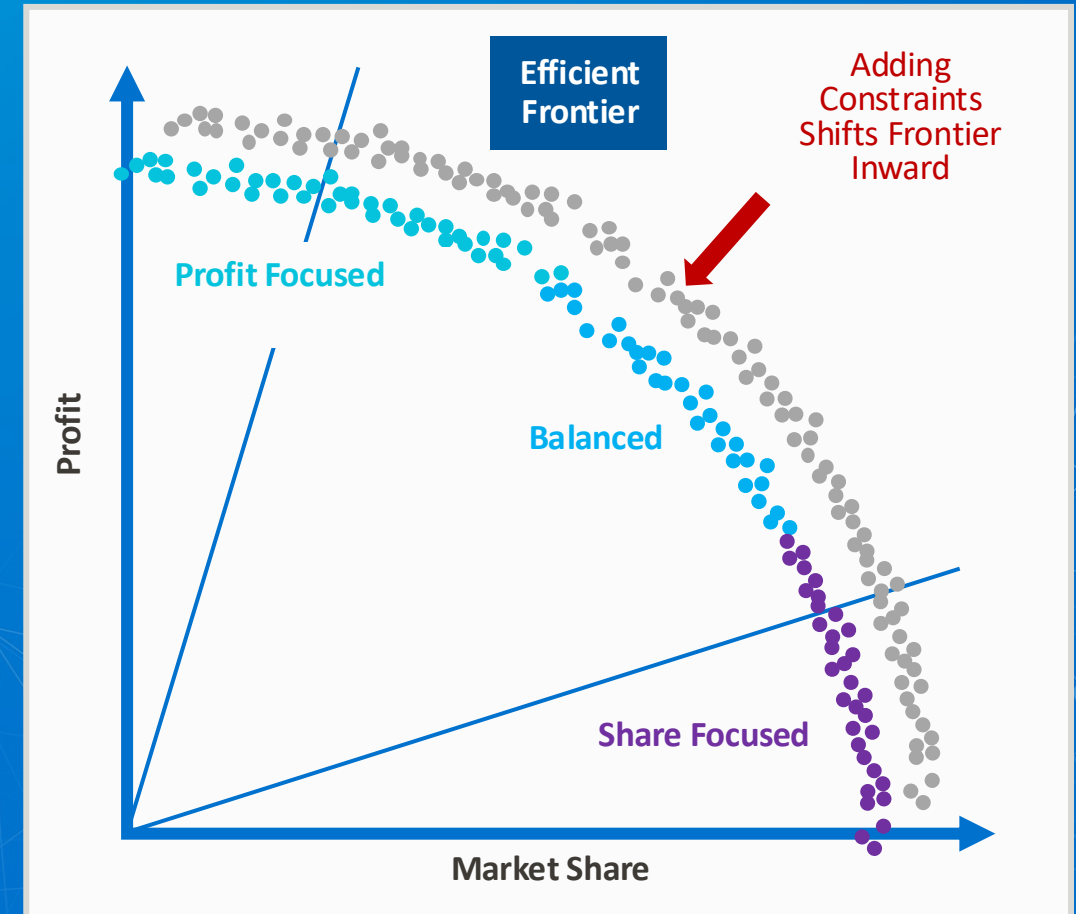
We use a **hybrid algorithm**, leveraging multiple methods and **custom heuristics** to generate **an efficient frontier**

Analysts Iterate with Vehicle Program Team to Refine the Constraints on the Desired Order Guide

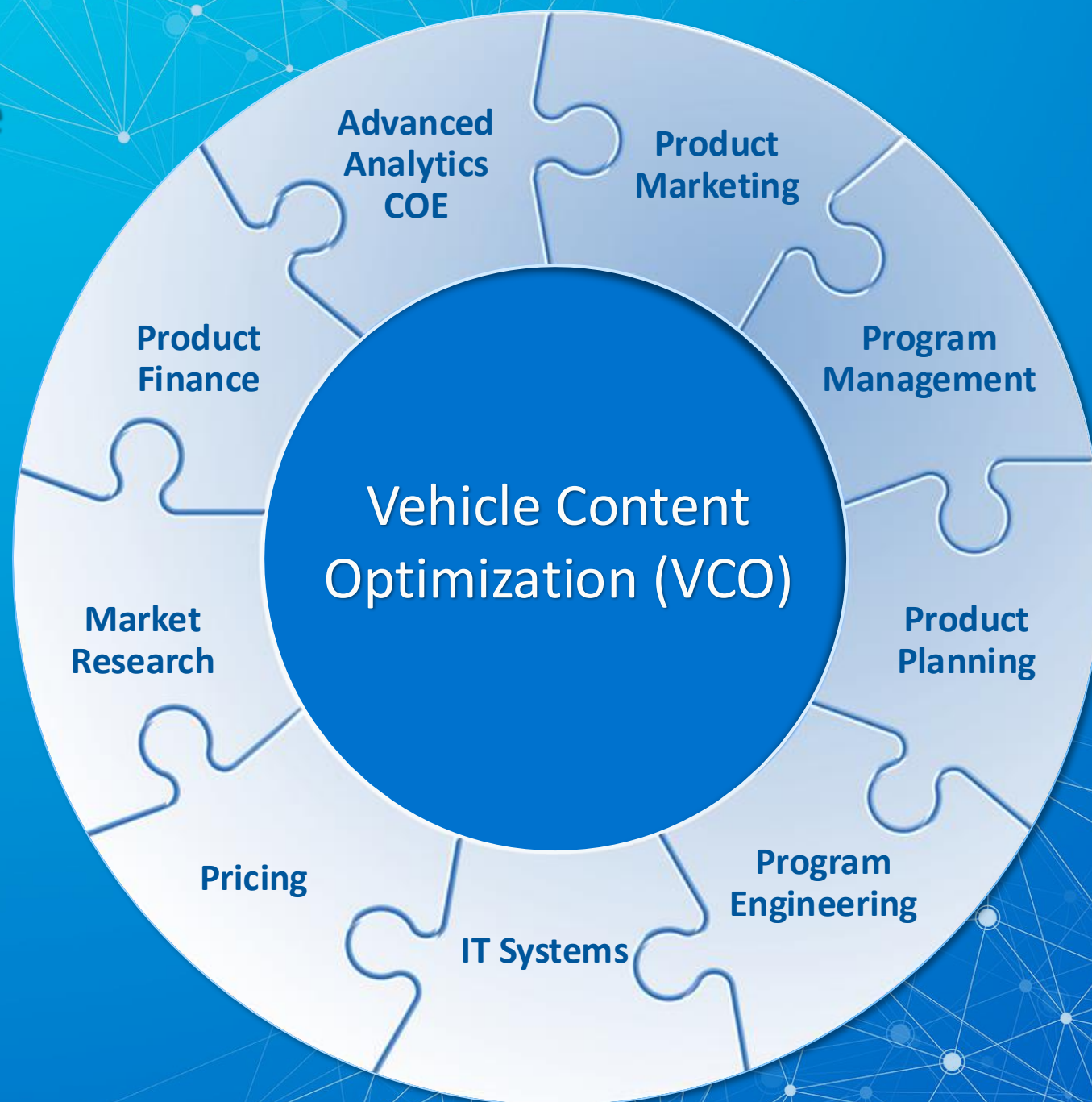


- Generate thousands of Order Guides scored on Profit & Market Share
- Organize the emergent recommendations for SME discussion
- Iterate with the program team to impose appropriate business constraints

CONSTRAINED ORDER GUIDES



It Takes a Village



Our Journey



Guided Analytics:
Gaining Traction

GEN 2:
2011 – 2016

Order Guide Optimization:
Full Business Integration

GEN 3:
2017 – Present

Proof of Concept:
Limited Business Support

GEN 1:
2008 – 2010

1

2

3



Vehicle Content Optimization (VCO)



Has been used on over
vehicle programs globally

85

Has enabled **\$4.4 Billion**
of lifecycle profit, as verified by GM Finance